Use Cases & End-to-End Journeys

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> **Rev 2 (2026-07-04, independent gap-analysis pass).** Added **UC-08 — Gateway failover**: the Architecture-Refined source (`04_ARCH §8`) sketches this flow and `HVPN-NFR-206`/`D-GW-SELECT` (decision register) depend on it, but no dedicated use-case + sequence diagram existed in this document (the other five core journeys named by the gap-analysis brief — connector onboarding, client connect+reach, policy-change propagation, and device revoke — were already covered: UC-05, UC-01, UC-04, and `personas-and-roles.md` §6.2 respectively). §1's catalogue table and §10's traceability table are extended to match.

> **Document role.** This is the Volume 1 (Product & Requirements) nano-detail spec enumerating HelixVPN's primary **end-to-end use cases**, each rendered as a step-by-step **journey** with a Mermaid sequence diagram, explicit preconditions and postconditions, and the exact **components and APIs** exercised. It turns the three roles, seven principles, and parity matrix of the Volume 1 overview (`00-product-scope-and-principles.md`) into runnable narratives that the §11.4.169 test types and the 8-criteria MVP DoD trace onto.
>
> **Document is a SPEC.** It describes *what the user does and what the system does in response*; it does not build the product. Each journey is a candidate e2e / full-automation test scenario (§11.4.98 — fully self-driving, no manual step after start) and is written so its postconditions are captured-evidence assertions, not prose claims (§11.4 / §11.4.69).

> **Evidence base.** Cross-references the Volume 1 overview (roles §3, two-way / multi-network §4, parity matrix §9, scope §10, decisions D1/D4/D6), the coordinator streaming contract (`v03-control-plane/svc-coordinator.md` — `WatchNetworkMap`, the < 1 s convergence + revoke SLOs, the reaction table), the telemetry presence + audit model (`v03-control-plane/svc-telemetry.md`), the no-logging invariant (`v05-security/no-logging-as-code.md`), and the transport escalation ladder (`v02-data-plane/transport-selection-ladder.md`). Research is cited by the overview's ids (`[04_ARCH §N]`, `[04_P1]`). Anything unproven is marked `UNVERIFIED` per §11.4.6.

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1. Use-case catalogue & actor model

The actors are the three roles of the overview plus the human personas (`00-...` §3, §5).

Actor	Role / persona	App
End user	Mullvad-style privacy/lab user	Helix Access
Network operator	exposes a LAN safely	Helix Connector (daemon)
Admin	manages tenants/users/devices/policy	Helix Console
Gateway control plane	Go (identity/registry/ipam/pki/policy/coordinator/events/telemetry/api)	server
		server

Actor	Role / persona	App
Gateway edge	Rust data plane (transports + WireGuard)	

UC	Title	Primary actor	Phase	Parity / differentiator
UC-01	Consumer first connect (privacy exit)	End user	MVP	F1, F7, F14
UC-02	Switch exit / server	End user	MVP	F1, X1
UC-03	Enable multihop	End user	P2	F10
UC-04	Admin enrolls a device & sets policy	Admin	MVP	F15, F17, X1
UC-05	Connector advertises a route	Network operator	MVP	X1, X2
UC-06	Censorship-evade via transport escalation	End user	MVP	F2, F6, F7, F8
UC-07	Kill-switch trips on network drop	End user	MVP	F11, F13
UC-08	Gateway failover	Gateway control plane (system-initiated)	P2	X3, NFR-206

Each journey below maps to the MVP Definition-of-Done acceptance criteria of the overview §10.2 where applicable, and each is a §11.4.98-compliant fully-automatable e2e scenario (no human step after start; non-introspectable UI falls back to the §11.4.117 pixel oracle).

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2. UC-01 — Consumer first connect (privacy exit)

Goal. A new end user installs Access, enrolls anonymously (no email), and gets a full-tunnel privacy exit to the internet — the Mullvad use case.

Preconditions.

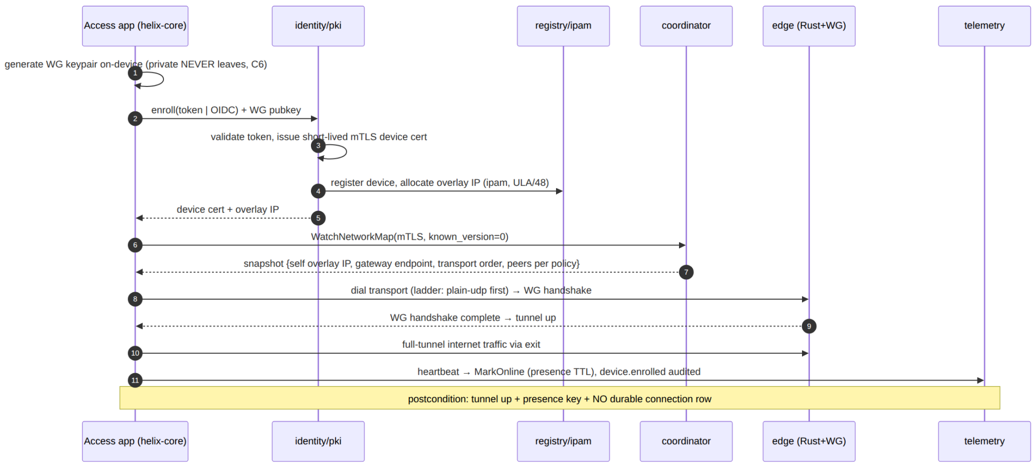
- A reachable Gateway (self-hosted or managed) with a public endpoint.
- An enrollment token (anonymous device-token path, no PII) OR an OIDC login.
- Access app installed on a supported platform (iOS/Android/Linux for MVP).

Postconditions (captured-evidence assertions).

- The device holds an overlay IP and a short-lived mTLS device cert.
- A WireGuard tunnel is established; the chosen exit carries the device's internet traffic (full-tunnel).
- A `presence:{tenant}:{device}` key exists in Redis (TTL'd); **no** durable connection row exists (NFR-300 / no-logging).
- An `device.enrolled` audit row exists (control action); **no** traffic audit.

Components / APIs exercised. `identity` (enroll), `pki` (device cert), `registry` (device + overlay-IP), `ipam` (overlay IP), `coordinator` (`WatchNetworkMap`), `helix-core` (tunnel + ladder), gateway `edge` (transport

- WG), `telemetry` (presence + audit).



Honest boundary (§11.4.6). On iOS the tunnel runs inside the NE memory ceiling (NFR-500) — first-connect on iOS is gated by Phase-0 G3 and is `UNVERIFIED` until that on-device capture exists. The journey's logic is platform-neutral; the iOS resource fit is the open risk, not the flow.

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3. UC-02 — Switch exit / server

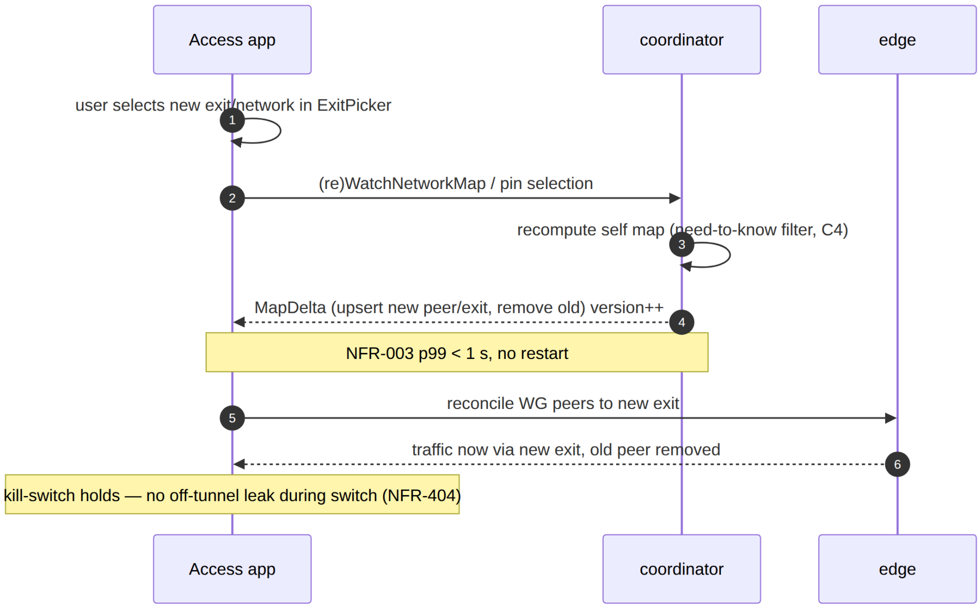
Goal. A connected user picks a different exit (privacy server) or a different joined network, and traffic re-routes without dropping the app session.

Preconditions. UC-01 complete (device connected, has overlay IP + cert). The target exit/network is one the device's policy grants (need-to-know, C4).

Postconditions.

- The new exit/network carries traffic; the old one no longer does.
- Convergence to the new route is **< 1 s** (NFR-003) via a `MapDelta`, with **no** client restart (DoD-5 spirit).
- The kill-switch never opens a leak during the transition (NFR-404).

Components / APIs. Access UI (`helix-ui` ExitPicker), `helix-core` (re-pin/reconnect), `coordinator` (delta), `edge` (re-route), `policy` (VisibleTo / AllowedIPs).



> **Need-to-know enforcement (C4).** The user can only switch to an exit/network their compiled policy grants — a non-granted target never appears in the snapshot (NFR-402), so the UI cannot offer it. This is the multi-network (X1) differentiator surfacing as a routing choice.

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4. UC-03 — Enable multihop

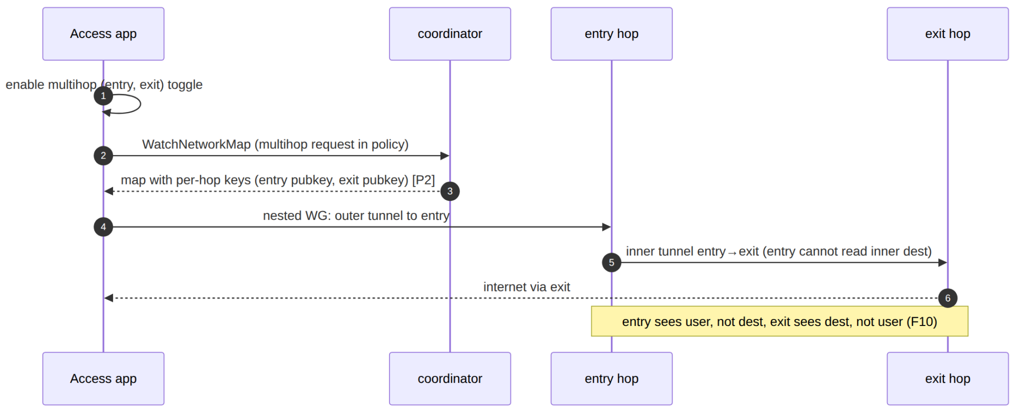
Goal. A privacy-conscious user enables entry/exit separation (multihop) so the entry node never sees the exit destination and vice-versa (parity F10). Phase 2.

Preconditions. P2 multihop available; user connected (UC-01). The tenant's policy/topology offers ≥2 hops in distinct jurisdictions.

Postconditions.

- Traffic traverses nested WireGuard (entry → exit), control-plane orchestrated.
- No single hop holds both the user identity and the destination.
- Latency rises (extra hop) within the obfuscation-aware budget (NFR-007) — exact added latency `UNVERIFIED` (P2 benchmark owns it).

Components / APIs. Access UI (multihop toggle), `helix-core` (nested WG), `coordinator` (per-hop keys in the map), `policy` (hop authorization), `edge`.



> **Phase boundary (§11.4.6).** Multihop is explicitly OUT of MVP scope (overview §10.2 → P2). This journey is the P2 contract; its postconditions are `UNVERIFIED` until the P2 nested-WG implementation and its e2e capture exist.

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5. UC-04 — Business admin enrolls a device & sets policy

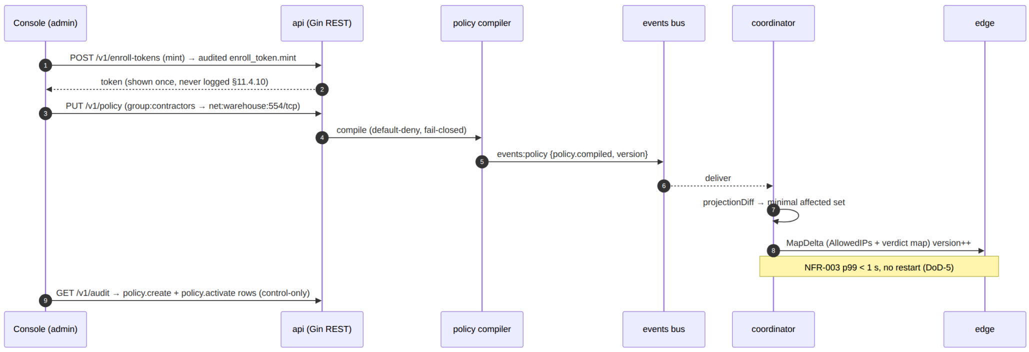
Goal. An admin (Console) creates a user/device enrollment, then writes a default-deny ACL granting a group access to a specific network/port — reflected on the edge in **< 1 s** with no restart.

Preconditions. Admin authenticated to Console (RBAC `admin/operator`). A tenant exists; at least one connector has advertised a network (UC-05).

Postconditions.

- An enrollment token is minted (audited `enroll_token.mint`); the device enrolls (UC-01) and appears in the registry.
- A policy edit (`group:contractors → net:warehouse:554/tcp`) compiles to per-peer `AllowedIPs` + an edge verdict map and reaches affected agents in **< 1 s** (NFR-003 / DoD-5).
- Audit holds `policy.create` + `policy.activate` control rows; **no** traffic.

Components / APIs. Console (`helix-ui` admin + REST `/v1/...`), `api` (Gin REST + WS/SSE), `identity` (enroll token), `policy` (compiler), `coordinator` (fan-out), `telemetry` (audit).



> **Authorization boundary.** `GET /v1/audit` requires `admin/operator`; `member` is denied; RLS is the floor so even a mis-scoped query returns only the caller's tenant rows (NFR-408). The minted token is shown once and never logged (§11.4.10).

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6. UC-05 — Connector advertises a route

Goal. A network operator installs the Connector daemon inside a private LAN; it dials **outbound** to the Gateway and advertises the LAN's CIDRs — no inbound port-forward, ever (P3 / X2).

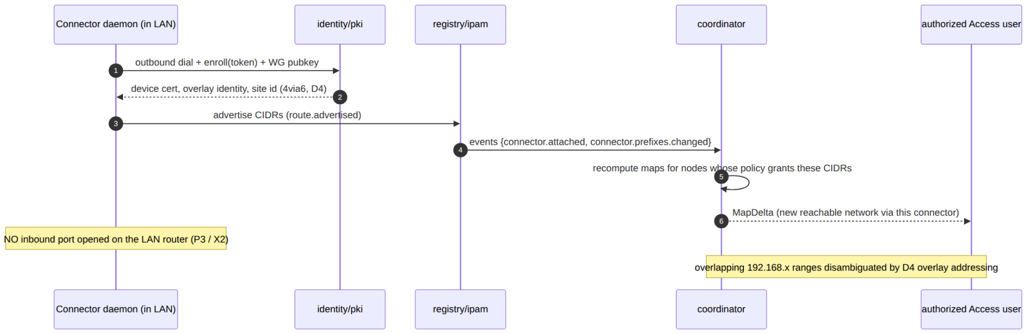
Preconditions. Connector daemon installed on a host inside the LAN. An enrollment token for the connector. Outbound reachability to the Gateway.

Postconditions.

- The connector holds a device cert + overlay identity and an attached `site` (4via6 site id, D4).

- Its advertised CIDRs are registered; authorized users' maps gain reachable peers/routes for that network.
- Overlapping RFC1918 ranges across connectors are disambiguated by the overlay addressing scheme (ULA /48 + 4via6 / per-network NAT fallback, D4).
- **No** inbound port was opened on the LAN's router (P3 assertion).

Components / APIs. Connector (`helix-core` advertise/route mode), `identity` /`pki`, `registry` (`route.advertised`), `ipam` (site), `coordinator` (reaction `connector.attached` / `connector.prefixes.changed`), `policy`.



> **Conflict handling (§11.4.6).** Two connectors advertising the same RFC1918 range raise `route.conflict.detected`; the coordinator does **not** push a delta for it — the conflict surfaces to Console and 4via6 site disambiguation resolves it (D4). The concrete addressing scheme is frozen in the Phase-1 IPAM design; until then the collision resolution detail is `UNVERIFIED`.

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7. UC-06 — Censorship-evade via transport escalation

Goal. A user on a censored network where plain WireGuard/UDP is blocked automatically escalates the transport ladder until an obfuscated rung (MASQUE/QUIC) carries the WG handshake — Mullvad's "automatic obfuscation" (F7), looking like ordinary web traffic.

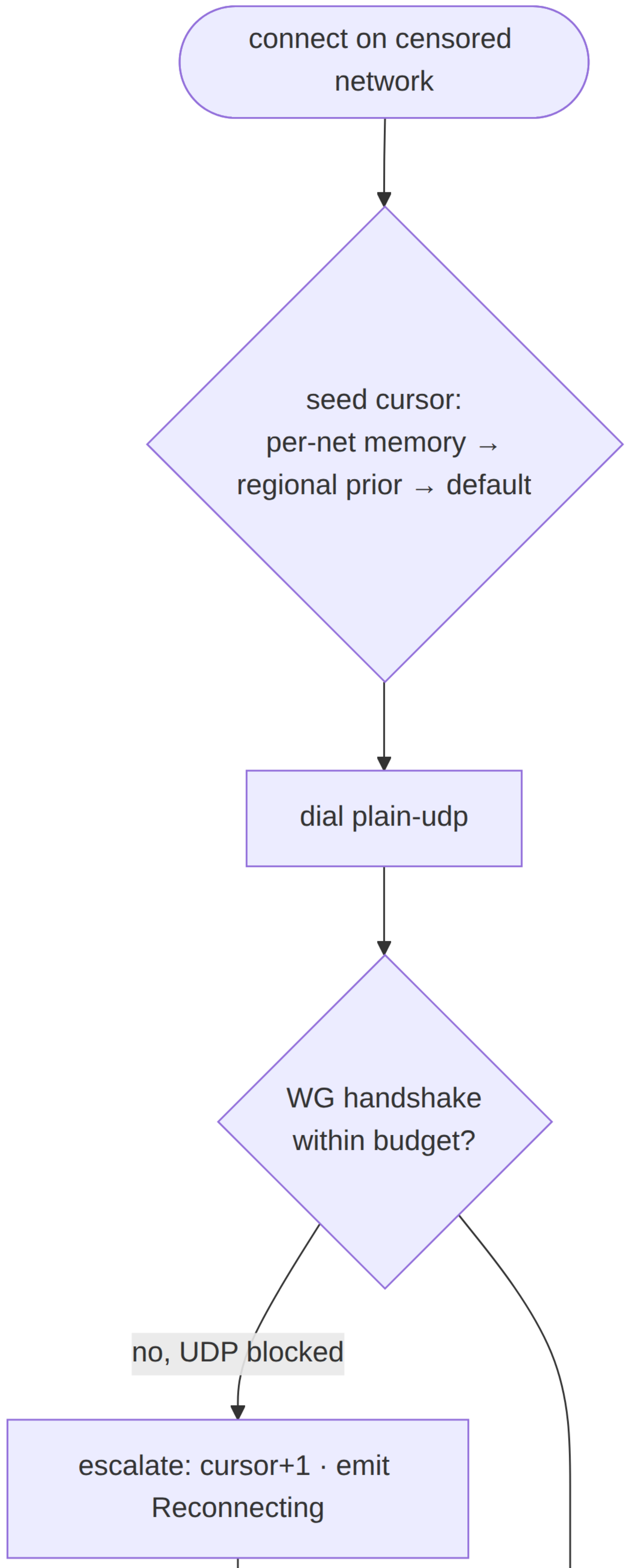
Preconditions. Device enrolled (UC-01). The current network blocks plain-UDP WG (DPI / UDP block). The ladder order is the default or a coordinator-pushed regional prior.

Postconditions.

- The ladder advances monotonically within the attempt (L-I8) until a rung completes a **WG handshake** (carrier-connect alone is not success, I2).

- The working rung is remembered per-network (fingerprint hash + rung index, **no** SSID plaintext, **no** endpoint, no per-connection log — I5).
- Aggregate escalation telemetry is recorded (counts only, no per-user flow).
- Throughput on the obfuscated rung meets the DPI-survival bar (NFR-002 ≥ 50 % of plain — TARGET `UNVERIFIED` until G2).

Components / APIs. `helix-core` ladder (`ladder.rs`), `helix-transport` (plain-udp / lwo / masque-h3 / …), `edge` (transport termination), `coordinator` (`TransportPolicy.order` regional prior), `telemetry` (aggregate ladder outcomes).


connect on censored network
seed cursor:
per-net memory →
regional prior → default
dial plain-udp
WG handshake within budget?
no, UDP blocked
escalate: cursor+1 · emit Reconnecting

> **No-logging on the evade path (§11.4.6).** Per-network memory persists only a fingerprint *hash* + last-good rung index — never the SSID, never the server endpoint, never a timestamped per-connection record (I5). The escalation event trace IS the captured evidence (L-I9), and it carries no destination. The ≥ 50 % DPI-survival throughput is a TARGET pending Phase-0 G2.

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8. UC-07 — Kill-switch trips on network drop

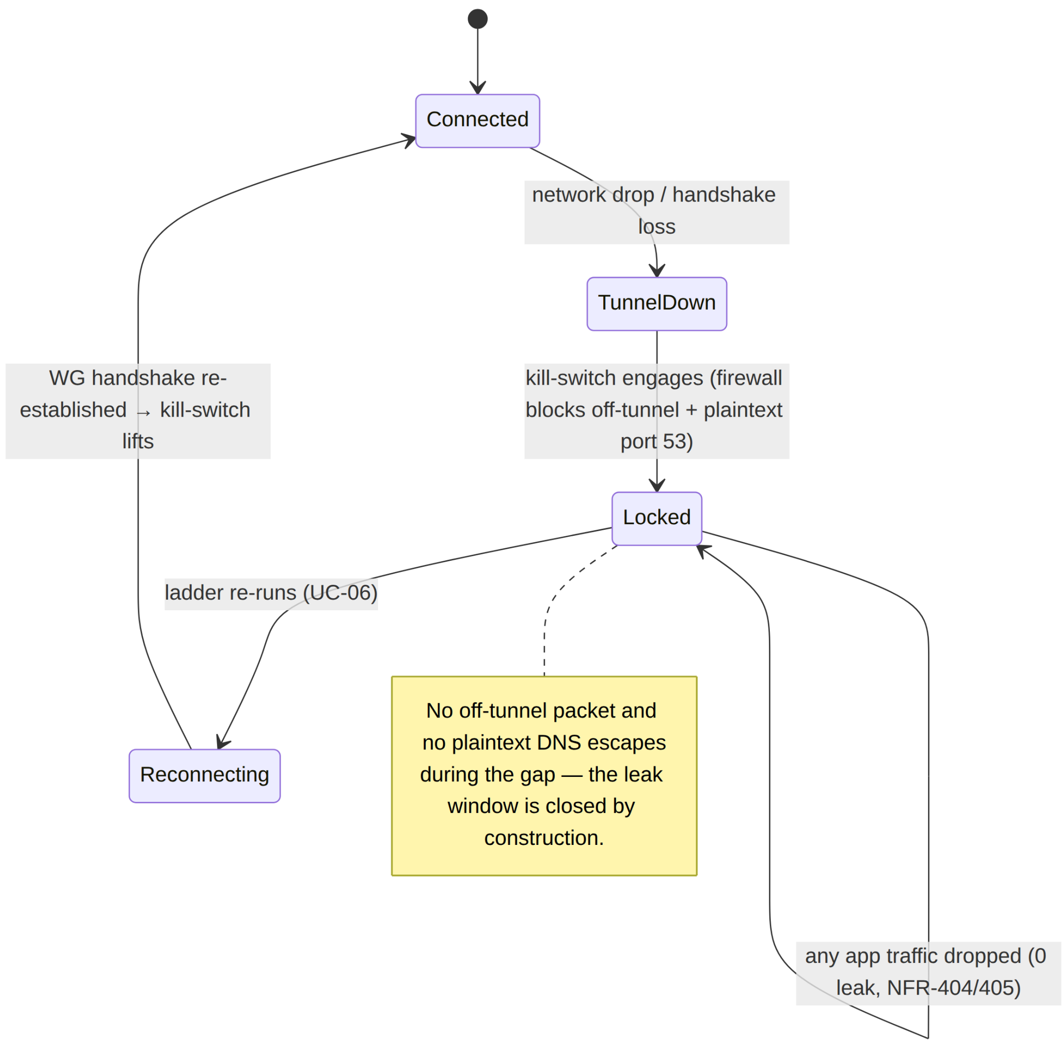
Goal. The user's underlying network drops (or the tunnel fails); the kill-switch blocks all off-tunnel traffic so nothing leaks, and DNS stays forced through the tunnel (F11 + F13).

Preconditions. Device connected with kill-switch enabled (default-on for the privacy posture). The OS firewall integration is active (`helix-core` state machine + per-OS firewall driver).

Postconditions.

- On tunnel-down, **0** packets traverse the physical interface off-tunnel (NFR-404 leak test).
- **0** plaintext DNS queries escape off-tunnel (NFR-405).
- On reconnect (ladder re-runs, UC-06), the kill-switch lifts only once a tunnel is re-established — never a leak-window during the gap.

Components / APIs. `helix-core` (kill-switch + DNS state machine), per-OS firewall shim (nftables/WFP/NE rules), `coordinator` (reconnect map), `edge`.



> **Verification (§11.4.69).** The kill-switch postconditions are proven by a packet-capture leak test on the physical interface during a forced tunnel-drop: a non-zero off-tunnel packet count is a FAIL. The capture IS the evidence; "no leak observed" without the capture is a PASS-bluff (§11.4).

8a. UC-08 — Gateway failover

Goal. A gateway (region/node) becomes unhealthy; the system re-points every affected agent (clients and connectors) to a healthy gateway endpoint with the selected transport preserved and without the operator manually touching any device — the multi-region HA promise (X3, `04_ARCH §8` "Gateway failover" sketch) made into a falsifiable journey.

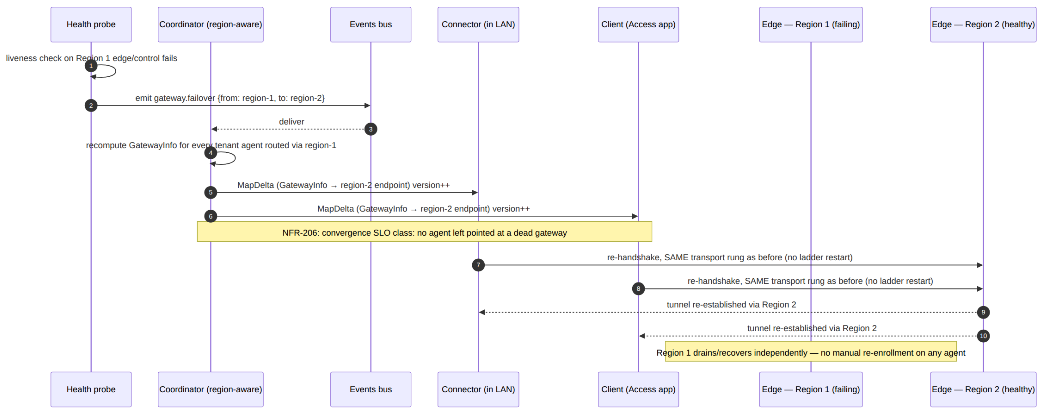
Preconditions. Phase-2 multi-region deployment: ≥2 gateway endpoints share control-plane state (stateless coordinators + Patroni Postgres / NATS JetStream, per `v06-deploy/ha-and-`

`multiregion.md`); a health probe monitors each gateway; `D-GW-SELECT` (decision register) has picked a selection mechanism (geoDNS default, anycast, or map-driven `GatewayInfo`).

Postconditions (captured-evidence assertions).

- The health probe's failure is observed and a `gateway.failover {from, to}` event is emitted — never a silent gap (`UNVERIFIED` exact probe-timeout value until the Phase-2 HA runbook fixes it).
- Every affected agent receives a `MapDelta` carrying an updated `GatewayInfo` (new endpoint) within the convergence SLO (NFR-003 class), **without** the agent needing a fresh enrollment or the user taking any action.
- The client/connector re-handshakes to the new endpoint with the **same selected transport** (ladder position preserved — no re-run of the escalation ladder from scratch) per NFR-206.
- No private-network inbound port is opened by the failover (P3 still holds — the connector still only ever dials outbound, now to the new endpoint).
- The region-failover RTO/RPO budget is the named target owned by `v06-deploy/disaster-recovery.md` — this journey's postconditions are the *functional* correctness bar; the *time* bound is `UNVERIFIED` until that runbook's drill produces a capture (§11.4.6 — no fabricated number here).

Components / APIs exercised. health-probe/liveness (`v06-deploy/observability.md`), `coordinator` (recomputes `GatewayInfo` per affected tenant, emits `gateway.failover`), events bus, `helix-core` (both Client and Connector capture modes — reconnect-preserving-transport), edge (new region terminates the re-established tunnel).



> **Phase + honesty boundary (§11.4.6).** UC-08 is explicitly **P2** (multi-region HA is out of MVP scope per the overview §10.2/§10.3) — Phase-1 self-host is single-gateway and has no failover target by design. The *mechanism* (health probe → `gateway.failover` event → `GatewayInfo`-only delta) is fixed here so `svc-coordinator.md` and `ha-and-multiregion.md` implement

> against one contract; the concrete probe-timeout and RTO/RPO numbers remain `UNVERIFIED` pending the Phase-2 DR drill (`v06-deploy/disaster-recovery.md`).

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9. Cross-cutting postcondition invariants

Every journey above, regardless of actor, must leave the system in a state that satisfies these invariants — they are the union of the seven principles and the no-logging guarantee surfacing as per-journey postconditions:

Invariant	Holds in every journey because	Verified by
Fail-static (P1)	the edge forwards from a cached verdict map; a control-plane blip during any journey never drops an established tunnel	chaos (kill control plane mid-journey)
Outbound-only (P3)	every actor *dials out*; the gateway is the only public listener	security (no-inbound assertion)
Need-to-know (C4)	every map a journey delivers is policy-filtered; a non-granted peer never appears	meta-test (skip-filter mutation FAILs)
< 1 s convergence (P4)	every map change in a journey is a minimal `MapDelta`	performance (`helix_reconcile_seconds`)
No-logging (P7)	no journey writes a durable connection/traffic row; presence is ephemeral; audit is control-only	security (schema-lint, deployed DB)
Anonymous identity (F15)	first-connect (UC-01) needs no email/SSO	integration (anonymous enroll)

> **Anti-bluff coupling (§11.4 / §11.4.98).** Each journey is authored as a fully self-driving e2e scenario: no human action after start, deterministic across re-runs (§11.4.50), captured evidence per postcondition. A journey that "passes"

> without its postcondition captures (presence key present, no durable row, delta on wire, zero leak) is a PASS-bluff at the use-case layer.

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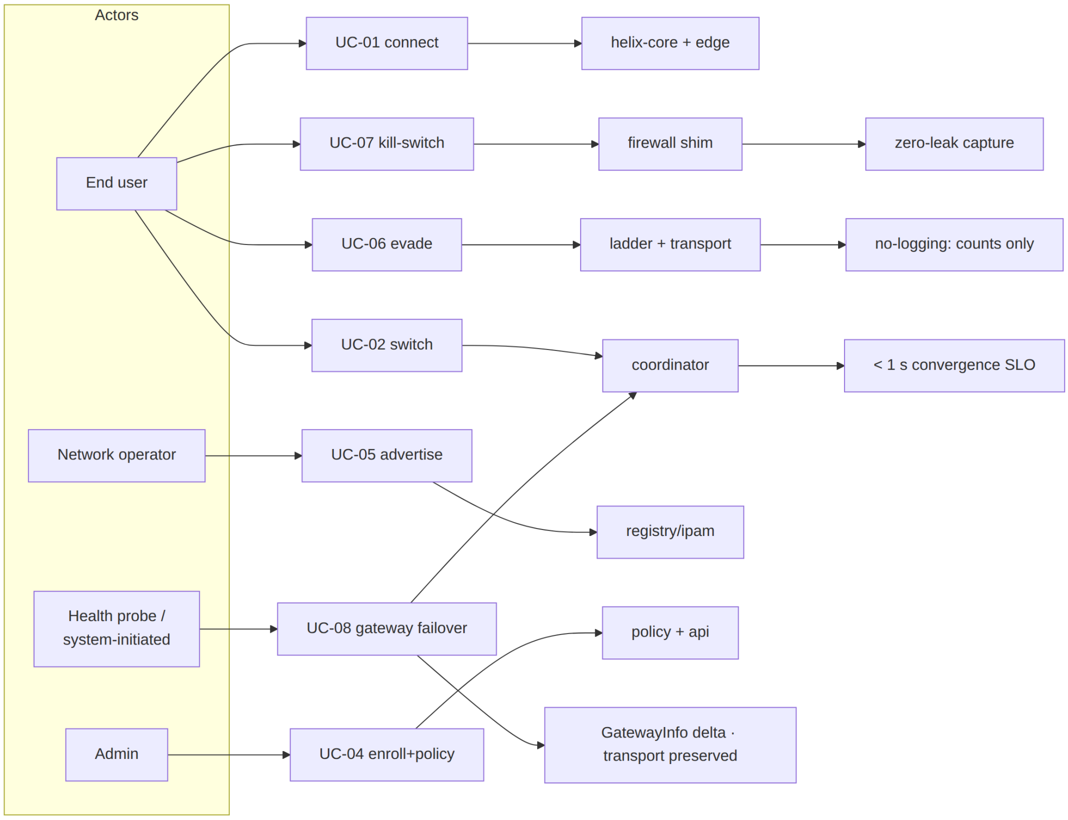
10. Use-case → component → API → test traceability

UC	Components	Key APIs / streams	DoD / NFR	§11.4.16 test typ
UC-01	identity, pki, registry, ipam, coordinator, core, edge, telemetry	enroll; `WatchNetworkMap`; transport dial	DoD-1/2, NFR-307/304	e2e / full automati
UC-02	core, coordinator, edge, policy	`WatchNetworkMap` delta	DoD-5, NFR-003/402	e2e
UC-03	core, coordinator, edge, policy	map w/ per-hop keys (P2)	F10, NFR-007	e2e (P2)
UC-04	api, identity, policy, coordinator, telemetry	REST `/v1/...`; events:policy; audit	DoD-3/5, NFR-003/408	e2e + integrati
UC-05	core, identity, pki, registry, ipam, coordinator, policy	`route.advertised`; `connector.attached`	X1/X2, NFR-103/104	e2e + integrati
UC-06	core (ladder), transport, edge, coordinator, telemetry	ladder; `TransportPolicy.order`	DoD-4, NFR-002/410/006	e2e (DPI sim)
UC-07	core (kill-switch/ DNS), firewall shim, coordinator, edge	firewall state machine	DoD-7, NFR-404/405	security , e2e (leak test)
UC-08	health-probe, coordinator, events bus, core (client+connector), edge (2 regions)	`gateway.failover` event; `GatewayInfo`-only `MapDelta`	X3, NFR-206 (P2)	chaos + integrati (region-failover drill)



> **Phase honesty (§11.4.6).** UC-01/02/04/05/06/07 are MVP-scoped; UC-03 (multihop) and UC-08 (gateway failover) are P2. P2/P3 postconditions are `UNVERIFIED` until their phase's implementation and captured e2e evidence exist — they are the forward contract, never asserted as already working.

Sources verified

- `docs/research/mvp/final/00-product-scope-and-principles.md` — three roles + control relationships (§3), two-way / multi-network exact meaning (§4), personas + app classes (§5), Mullvad-parity matrix F1–F17 + X1–X5 (§9), phase scope + 8-criteria MVP DoD (§10), decisions D1 (MASQUE)/D4 (ULA+4via6)/D6 (asymmetric per-leg) (§11).
- `docs/research/mvp/final/v03-control-plane/svc-coordinator.md` — the `WatchNetworkMap` streaming handler, `known_version` resume, the reaction table (`device.enrolled/device.revoked/connector.attached/ connector.prefixes.changed/policy.compiled`), need-to-know `VisibleTo` filtering (C4), the < 1 s convergence + < 1 s revoke SLOs (§7).
- `docs/research/mvp/final/v03-control-plane/svc-telemetry.md` — ephemeral Redis presence (MarkOnline/Refresh/Offline, TTL), the control-action audit vocabulary
  - meta-shape guard, counts-only metrics.
- `docs/research/mvp/final/v05-security/no-logging-as-code.md` — the no-logging postcondition invariants (no durable connection

row, presence ephemeral, audit control-only) every journey must satisfy.

- `docs/research/mvp/final/v02-data-plane/transport-selection-ladder.md` — the escalation ladder behind UC-06 (seed order, monotonic cursor L-I8, WG-handshake success criterion I2, per-network memory + no-logging I5, aggregate telemetry).
- `04_VPN_CLD/HelixVPN-Architecture-Refined.md` §8 ("Gateway failover" sequence sketch) and `docs/research/mvp/final/v06-deploy/ha-and-multiregion.md` + `disaster-recovery.md` (region-failover mechanism + the RTO/RPO budget owner) — the source for UC-08.

Constitution bindings applied: §11.4.44 (revision header), §11.4.6 (no-guessing — P2/P3 postconditions + unproven targets marked `UNVERIFIED`), §11.4.98 (every journey authored as a fully self-driving e2e scenario, no manual step after start), §11.4.50 (deterministic re-run), §11.4.69/§11.4.107 (each postcondition is a captured-evidence assertion), §11.4.117 (pixel-oracle fallback for non-introspectable UIs), §11.4.25 (use-case → component → test coverage ledger §10), §11.4.65/§11.4.153 (HTML+PDF[+DOCX] exports follow in refinement).